



Kubernetes beyond the basics

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Kubernetes internals

Agenda

- What components make up a Kubernetes cluster
- What each component does and how it does it
- How creating a Deployment object results in a running pod
- What a running pod is
- How the network between pods works
- How Kubernetes Services work
- How high-availability is achieved

Kubernetes cluster is split into two parts

- The Kubernetes Control Plane
- The (worker) nodes

Plus some add-ons components

Let's look more closely at what these two parts do and what's running inside them.

Components of the control-plane

- The etcd distributed persistent storage
- The API server
- The Scheduler
- The Controller Manager

These components store and manage the state of the cluster, but they aren't what runs the application containers.

Components running on the worker nodes

- The Kubelet
- The Kubernetes Service Proxy (kube-proxy)
- The Container Runtime (Docker, rkt, or others)

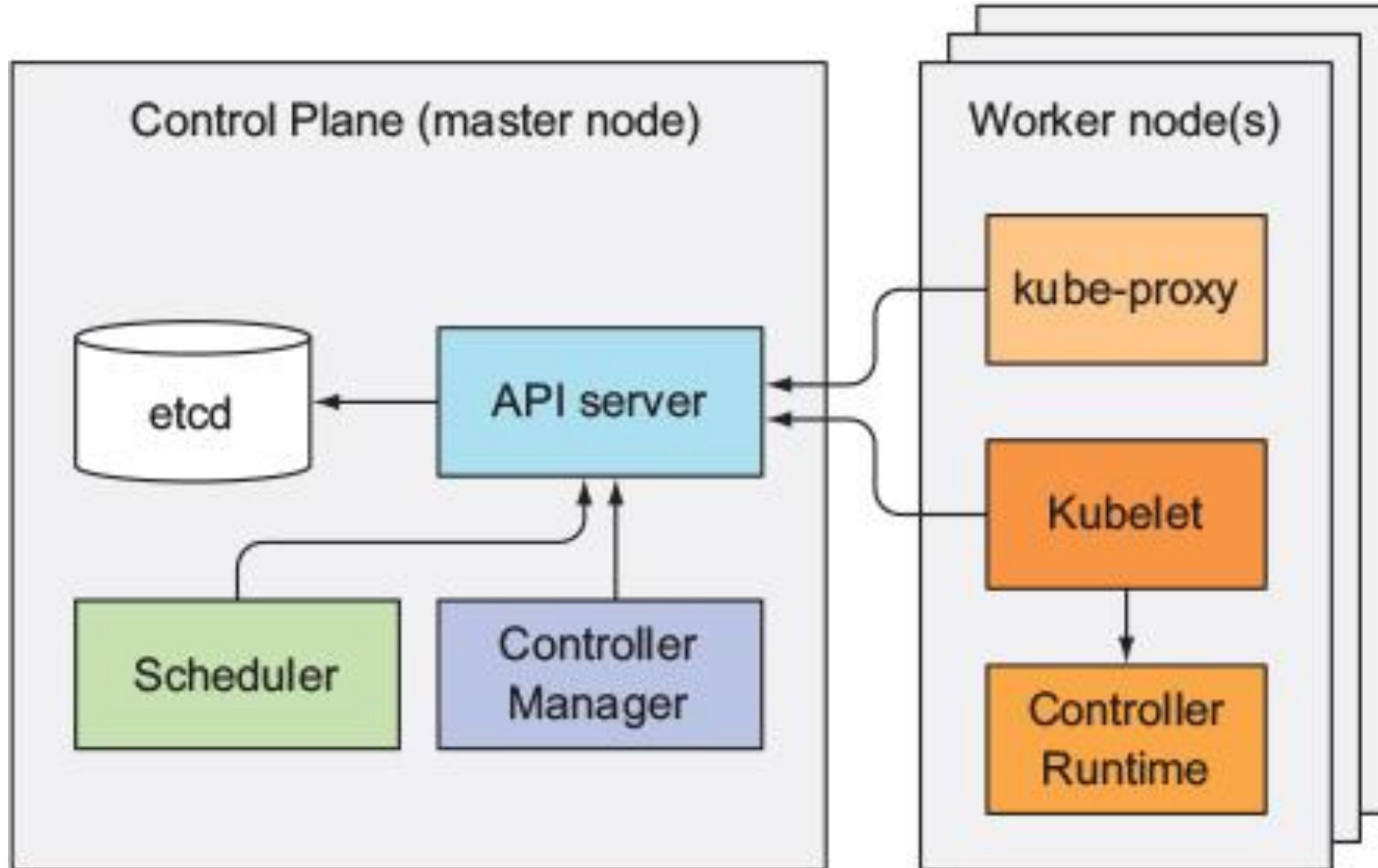
Add-on components

- The Kubernetes DNS server
- The Dashboard
- An Ingress controller
- metrics-server
- The Container Network Interface network plugin

And also:

<https://github.com/kubernetes/kubernetes/tree/master/cluster/addons/>

Components of the Control Plane and the worker nodes



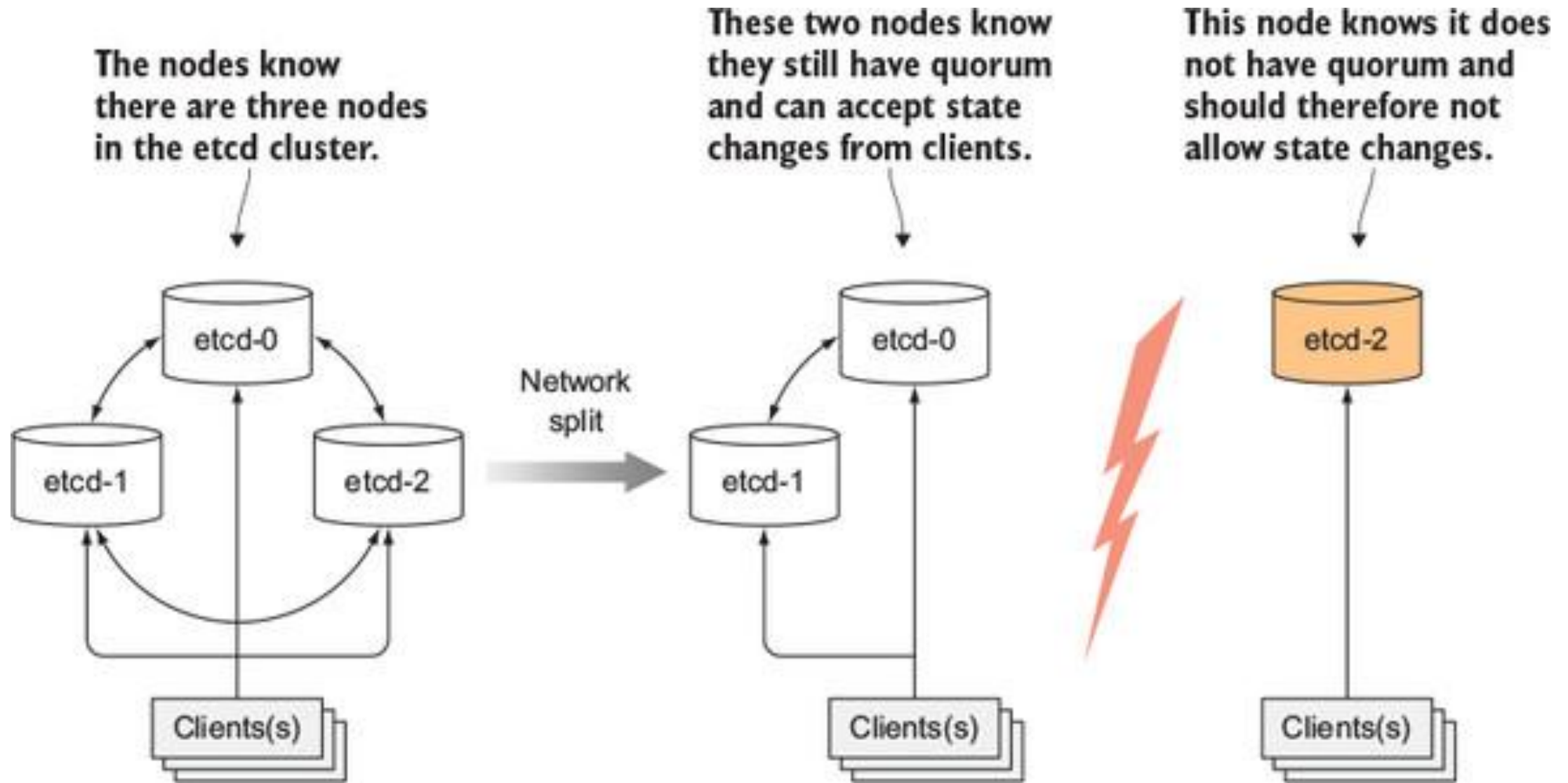
Check the status of control-plane components

```
$ kubectl get componentstatuses
```

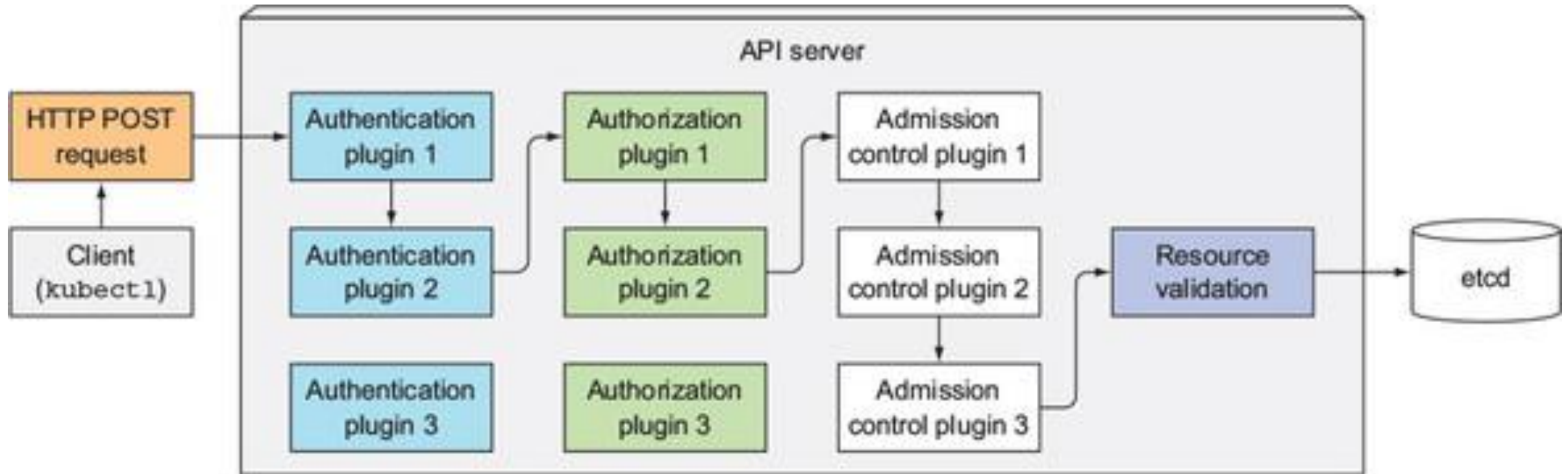
NAME	STATUS	MESSAGE	ERROR
scheduler	Healthy	ok	
controller-manager	Healthy	ok	
etcd-0	Healthy	{"health": "true"}	

Kubernetes system components communicate only with the API server. They don't talk to each other directly.

Components of the Control Plane and the worker nodes



Operations of the API server



Questions?

Merci!

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